

1.1 Atomic Structure

Question Paper

Course	CIEA Level Chemistry
Section	1. Physical Chemistry
Topic	1.1 Atomic Structure
Difficulty	Easy

Time allowed: 20

Score: /10

Percentage: /100

Question 1

In 1808 John Dalton published his atomic theory. This was the first complete attempt to describe matter in terms of atoms and contained four theories.

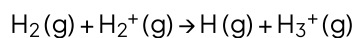
Which of his theories is still considered to be correct today?

- A. All matter is made of indivisible atoms.
- B. All atoms are very small in size.
- C. All the atoms of one element are different in mass from all the atoms of other elements.
- D. All atoms of the same element will have the same mass.

[1 mark]

Question 2

The reaction shown occurs in gas clouds throughout the Universe.



What is the atomic structure of the H_3^+ ion?

	protons	neutrons	electrons
A	3	0	1
B	3	0	2
C	2	1	1
D	2	1	2

[1 mark]

Question 3

Which row is correct?

	the atomic number is the number of	the proton number is the number of	the mass number is the number of
A	atoms	protons	protons
B	electrons	neutrons	nucleons
C	electrons	neutrons	protons
D	protons	protons	nucleons

[1 mark]

Question 4

The phosphide ion, $^{32}_{15}\text{P}^{3-}$, is used in medicine as a radioactive treatment for some forms of cancer.

What is the composition of the phosphide ion?

	protons	neutrons	electrons
A	15	17	32
B	15	17	18
C	17	15	15
D	17	15	32

[1 mark]

Question 5

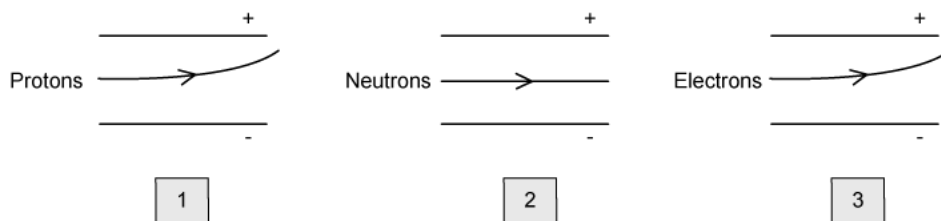
Which option correctly describes the relative charges and masses of the subatomic particles?

	proton	neutron	electron	the relative mass of an electron
A	+1	0	-1	1
B	0	+1	+1	$\frac{1}{1840}$
C	+1	0	-1	$\frac{1}{1840}$
D	0	+1	-1	1

[1 mark]

Question 6

The trajectories of subatomic particles shot through an electric field produced in a vacuum are shown in the diagrams.



Which diagrams are correct?

- A. 2 and 3
- B. 1 and 2
- C. 1, 2 and 3
- D. 2 only

[1 mark]

Question 7

Which row correctly describes the characteristics of the nucleus and surrounding space?

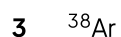
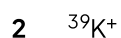
	Nucleus	Surrounding space
A	Small, dense and neutral	Mainly empty space
B	Large, dense and positive	Densely populated with electrons
C	Large, dense and neutral	Densely populated with electrons
D	Small, dense and positive	Mainly empty space

[1 mark]

Question 8

X is a particle with 19 electrons and 20 neutrons.

Three species are shown below.



Which of the three species could particle X be?

- A. 1 only
- B. 1 and 2
- C. 2 and 3
- D. 1, 2 and 3

[1 mark]

Question 9

Which of the following statements describes a free radical?

- A. A species with a lone pair of electrons
- B. A species with a half-filled d-orbital
- C. A species with one or more unpaired electrons
- D. A species with a partially filled outer shell

[1 mark]

Question 10

Which of the following statements describes first ionisation energy?

- A. The energy required to remove one mole of electrons from one mole of gaseous atoms.
- B. The energy required to remove the outermost electron from each atom in one mole of gaseous atoms.
- C. The energy required to remove the outermost electron from each atom in one mole of atoms.
- D. The energy required to produce one mole of ions from one mole of gaseous atoms.

[1 mark]

Model Answers: Easy

1

The correct answer is **B** because:

- atoms are very, very small with their sizes measured in **angstroms**, whereas 1 angstrom is equivalent to one ten-billionth of a metre (0.1 nanometres).

A is incorrect as atoms can be split into simpler parts (subatomic particles).

C is incorrect as the defining feature of an atom is the number of protons in its nucleus (atomic number); carbon-14 atoms exist, which have the same mass as a nitrogen atom.

D is incorrect as we know that isotopes exist – atoms of the same element with different masses (due to variation in the number of neutrons for an atom of a particular element).

2

The correct answer is **B** because:

- Each hydrogen atom contains only **1 proton** in its nucleus.
- There are 3 hydrogen atoms hence there are 3 protons.
- The ion has a single positive charge, so it must have 2 electrons since $3 - 2 = +1$.
- There are no neutrons in the hydrogen atom.

A, C & D are incorrect as these options do not correctly describe the composition of the H_3^+ ion.

3

The correct answer is **D** because:

- The atomic number is the same as the proton number and refers to the number of **protons** in the nucleus of an atom.
- The mass number is the number of protons and neutrons (**nucleons**) together in the nucleus.

A, B & C are incorrect as these options do not correctly describe the terms atomic number, mass number and proton number.

4

The correct answer is **B** because:

- The atomic number is 15 so there are 15 protons.
- There are $32 - 15 = 17$ neutrons.
- The charge is 3^- so it has gained 3 electrons, therefore there are $15 + 3 = 18$ electrons.

A, C & D are incorrect as these options do not correctly describe the composition of ${}_{15}^{32}\text{P}^{3-}$.

5

The correct answer is **C** because:

- Protons are **positively** charged, neutrons are **neutral** and electrons are **negatively** charged.
- The relative mass of protons and neutrons is 1, while that of an electron is $\frac{1}{1840}$, so one electron has 1840 times less mass than a proton or a neutron.

A, B & D are incorrect as only option **C** has correctly described the subatomic particles.

6

The correct answer is **A** because:

- In each diagram, the upper plate is **positively** charged and the lower plate is

negatively charged.

- In diagram 2 neutrons pass straight through **unaffected** as they have **no charge**.
- In diagram 3 **electrons** are attracted to the **positive** upper plate since they are **negatively** charged particles.

B, C & D are incorrect as only diagrams 2 and 3 correctly describe the paths that the subatomic particles follow. In diagram 1, the proton would be attracted to the negatively charged plate.

7

The correct answer is **D** because:

- The nucleus contains protons and neutrons which account for the mass of the atom since each one is 1840 times heavier than an electron.
- They are packed close together inside the nucleus, so it is a **small** and **dense** area which is **positively charged** due to the presence of the **protons**.
- The area surrounding the nucleus is populated with electrons which are tiny particles spread out in energy shells.
- In relative terms, the diameter of an atom is around 1×10^5 times larger than the diameter of a nucleus, hence the surrounding space is mainly empty.

A, B & C are incorrect as these options incorrectly describe the characteristics of the atom.

8

The correct answer is **A** because:

- $^{40}\text{Ca}^+$ has:
 - 20 protons (**atomic number** found from the periodic table).
 - An **atomic mass** of 40 (stated in the question).
 - $40 - 20 = 20$ neutrons.
 - A **+1 charge**, therefore it has one less electron than the Ca atom.
 - $20 - 1 = 19$ electrons.

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 - 20 protons (**atomic number** found from the periodic table).
 - An **atomic mass** of 40 (stated in the question).
 - $40 - 20 = 20$ neutrons.
 - A **+1 charge**, therefore it has one less electron than the Ca atom.
 - $20 - 1 = 19$ electrons.

B, C & D are incorrect as only statement 1 is correct. ^{38}Ar has $38 - 18 = 20$ neutrons but only 18 electrons. $^{39}\text{K}^+$ has $39 - 19 = 20$ neutrons and $19 - 1 = 18$ electrons

9

The correct answer is **C** because:

- Free radicals are atoms or groups of atoms that have one or more **unpaired electrons**.
- They are **highly reactive** due to the presence of the unpaired electron(s).
- An example is the **homolytic fission** of halogens by UV light:
 - $\text{Cl}_2 \rightarrow 2\text{Cl}\cdot$
- The dot after the Cl indicates the presence of an unpaired electron.
- Free radicals are different from ions because they do **not** have a charge.

A, B & D are incorrect as these statements do not match the description of a free radical.

10

The correct answer is **B** because:

- The first ionisation energy refers to the removal of the **outermost electron** from an atom in the **gaseous** state.
- It is a **molar** function so refers to the removal of one mole of electrons to form one mole of gaseous cations.

A, C & D are incorrect as these statements do not match the description of the first ionisation energy.